CFD Setup Instructions

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Contents

1 Geometry Creation in DesignModeler **3**

1.1 Setup . . . 3

1.2 The Main Pipeline . . . 3

1.3 The Branch Pipeline (The "T") . . . 4

1.4 Verification & Cleanup . . . 5

2 Parametrization of the Project **6**

2.1 Parametric Workbench & Geometry Setup . . . 7

2.1.1 Create Your Base Systems . . . 7

2.1.2 Open DesignModeler . . . 7

2.1.3 Build the Static Main Pipeline . . . 7

2.1.4 Create the Adjustable Angled Plane . . . 8

2.1.5 Sketch the Adjustable Branch . . . 8

2.1.6 Extrude the Branch . . . 8

2.1.7 The "Fluid" Fix & Cleanup . . . 9

2.2 Verifying the Setup . . . 10

3 Parametric Mesh Generation **11**

3.1 Pre-Requisite: Geometry BOI Setup . . . 11

3.2 Global Setup & Quality Controls . . . 11

3.3 Boundary Labels (Named Selections) . . . 12

3.4 Structured Core Algorithm . . . 12

3.5 Parametric Wake Refinement (Body of Influence) . . . 13

3.6 DDES Boundary Layer (Inflation) . . . 13

3.7 Automated Scaling Setup (Workbench Parameter Set) . . . 14

4 Literature Review **15**

4.1 The Foundation and The Problem . . . 15

4.1.1 The Big Picture Goal . . . 15

4.1.2 The Core Engineering Challenge . . . 15

4.1.3 The Mixing Problem (The Focus of the Paper) . . . 16

4.2 Model Description and Mathematical Framework . . . 16

4.2.1 The Physical Setup . . . 16

4.2.2 The Core Equations (Mixing & Concentration) . . . 17

4.2.3 The Computational Math (Fluid Dynamics) 18
4.2.4 Implementation and Validation 18
4.3 Governing Equations and Numerical Methods 19
4.3.1 The Governing Equations (The Laws of Physics) 19
4.3.2 Turbulence Modeling (Taming the Chaos) 20
4.3.3 Boundary Conditions (Setting the Stage) 20
4.3.4 Implementation and Validation (Building the Virtual Pipe) 20

Chapter 1

Geometry Creation in DesignModeler

This chapter uses the most foolproof method to create the T-junction fluid domain: two simple sketches and two extrusions—no offset planes, no confusing coordinates—resulting in one perfect, watertight fluid body.

1.1 Setup

1. In Workbench, double-click **Geometry** to open DesignModeler.
2. When prompted (or from the top menu), set your **Units** to **Millimeter**.

1.2 The Main Pipeline

We will draw the main pipe starting at the origin and extruding it backward.

1. In your **Tree Outline** (left side), click on **XYPlane**.
2. Click the **New Sketch** icon (the small blue drawing icon with a yellow star in the top toolbar). This creates `Sketch1`.
3. At the bottom of the Tree Outline, switch from the *Modeling* tab to the **Sketching** tab.
4. Go to **Draw** → **Circle**.
5. Click exactly on the **center origin point** (marked by the red/green axis intersection) and drag outward to draw a circle.
6. Go to **Dimensions** → **General**. Click on the edge of your circle to apply a dimension.
7. Look at the **Details View** (bottom left). Change the diameter (`D1`) to **460**.
8. Switch back to the **Modeling** tab at the bottom.
9. Click the **Extrude** button in the top toolbar.

10. In the Details View for the Extrude:
    - **Operation:** Add Material
    - **Direction:** Normal
    - **Extent Type:** Fixed
    - **Depth: 9200**
11. Click the yellow **Generate** button (lightning bolt) at the top.

Result: You now have a 9200 mm long cylinder lying along the Z-axis.

1.3 The Branch Pipeline (The "T")

Instead of making a new plane, we sketch on the standard plane that slices horizontally right through the middle of your main pipe.

1. In your Tree Outline, click on **ZXPlane**.
2. Click the **New Sketch** icon. This creates `Sketch2`.
3. Look at the 3D compass in the bottom right of your screen. Click the **Y-axis arrow**. This will turn your camera so you are looking straight down at the pipe, making it easy to draw.
4. Switch to the **Sketching** tab.
5. Go to **Draw** → **Circle**.
6. Hover your mouse over the horizontal axis line (the Z-axis). When you see a small **"C"** appear next to your cursor (this means "Coincident," meaning you are locked onto the line), click and drag to draw a circle anywhere along that line.
7. Go to **Dimensions** → **General**. Click on the edge of this new circle. Set the diameter (`D2`) to **115**.
8. Now we need to position it exactly in the middle. Go to **Dimensions** → **Horizontal** (or Vertical, depending on how your screen is rotated; you just want to measure the distance along the pipe).
9. Click the **center origin point** (where the main pipe started), and then click the **center point of your small circle**.

10. In the Details View, set this distance dimension to **4600**. (This centers it exactly at the midpoint of the 9200 mm pipe.)
11. Switch back to the **Modeling** tab.
12. With `Sketch2` selected in the tree, click **Extrude**.
13. In the Details View for the Extrude:
    - **Operation:** Add Material — *this is the magic step; it automatically melts the two pipes together into one fluid body.*
    - **Direction:** Normal — *this extrudes it straight up, forming the T.*
    - **Extent Type:** Fixed
    - **Depth: 1380** — *we use 1380 because your branch is 1150 mm long, plus the 230 mm radius of the main pipe it has to push through to get out.*
14. Click **Generate**!

1.4 Verification & Cleanup

1. Look at your Tree Outline. At the very bottom, it should say **1 Part, 1 Body**. Expand it.
2. Right-click on the "Solid" and select **Rename**. Name it `T_Junction_Fluid`.
3. If you right-click the body and select **Properties**, you will see it is recognized as a single continuous volume.

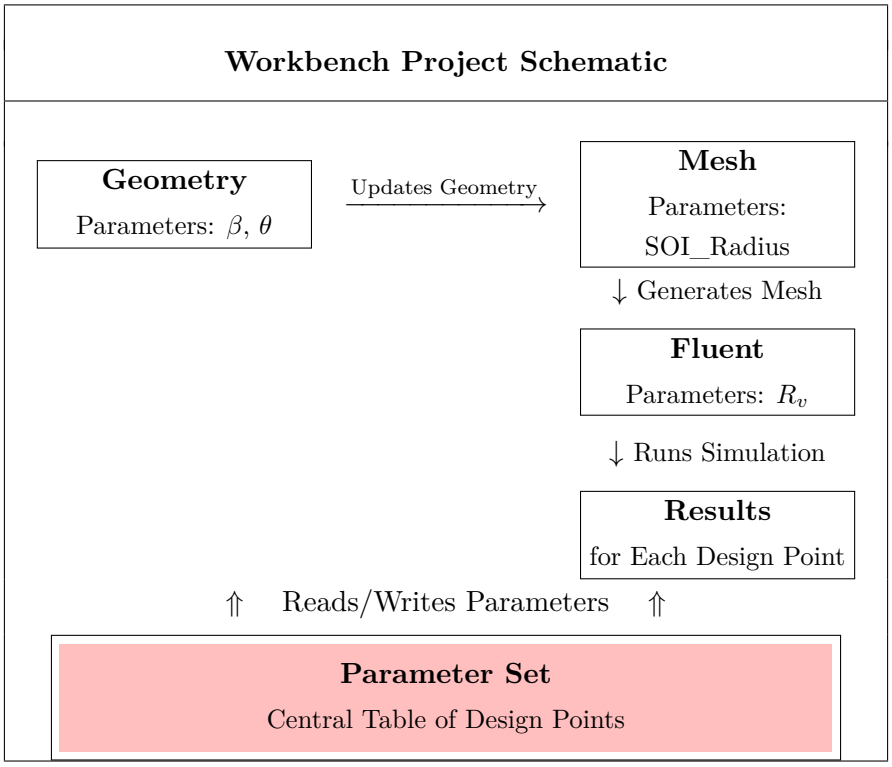
You now have the exact geometry required for your hydrogen mixing simulation. You can save the project, close DesignModeler, and move on to meshing!

Chapter 2

Parametrization of the Project

This chapter describes how to set up a fully parametric ANSYS Workbench project so that the geometry, mesh, and solver settings can all be driven from a single central table. The overall architecture is illustrated below.

Parametric Architecture Overview



The **Parameter Set** acts as the central hub: it reads and writes parameters to Geometry, Mesh, and Fluent. Each row in the Parameter Set table represents a unique *Design Point*—a specific combination of diameter ratio (β), injection angle (θ), and volumetric dilution (R_v). ANSYS will automatically update the geometry, regenerate the mesh, and re-run the solver for each design point.

2.1 Parametric Workbench & Geometry Setup

2.1.1 Create Your Base Systems

Because switching from top injection to bottom injection completely changes the physics and gravity vectors, we will make two separate bases.

1. Open ANSYS Workbench.
2. From the Toolbox on the left, click and drag **Fluid Flow (Fluent)** into the project schematic.
3. Rename this system to `Top_Injection_Base`.
4. Click and drag a *second* **Fluid Flow (Fluent)** system into the schematic right below it. Rename it to `Bottom_Injection_Base`.

Note: The rest of these steps apply to building the `Top_Injection_Base`*. You will repeat this for the bottom base later.*

2.1.2 Open DesignModeler

1. Right-click the **Geometry** cell of `Top_Injection_Base` and select **Properties**.
2. In the Properties pane on the right, ensure the **Parameters** box is checked (it usually is by default).
3. Double-click the **Geometry** cell to open DesignModeler.
4. Set your Units to **Millimeter**.

2.1.3 Build the Static Main Pipeline

1. In the Tree Outline, click **XYPlane**.
2. Click the **New Sketch** icon (creates `Sketch1`).
3. Switch to the **Sketching** tab → **Draw** → **Circle**.
4. Click the origin (0,0) and drag to draw a circle.
5. Go to **Dimensions** → **General**, click the circle, and set the diameter (`D1`) to **460**.
6. Switch back to the **Modeling** tab, click **Extrude**.
7. Set **Depth** to **9200**.

8. Click **Generate**.

2.1.4 Create the Adjustable Angled Plane

This is where we parameterize the Injection Angle (θ).

1. In the Tree Outline, click the **ZXPlane**.
2. Click the **New Plane** icon.
3. In the Details View for this new plane, apply these exact settings:
   - **Transform 1 (RMB):** Select **Offset Z**. Set the value to **4600**. *(This centers the plane.)*
   - **Transform 2 (RMB):** Select **Rotate about X**. Set the value to **90°**. *(Note: When you do the Bottom Injection base later, this will be $-90°$ or 270°.)*
4. **Parameterize it:** Right next to that `90°` value input, click the small, blank white square. A blue "P" will appear. Name the parameter `Inj_Angle`.
5. Click **Generate**.

2.1.5 Sketch the Adjustable Branch

This is where we parameterize the Diameter Ratio (β).

1. Select your newly created plane in the tree.
2. Click **New Sketch** (creates `Sketch2`).
3. Switch to the **Sketching** tab → **Draw** → **Circle**.
4. Click exactly on the origin of this new plane and draw a circle.
5. Go to **Dimensions** → **General**, and click the circle.
6. Set the diameter value to **115** (your starting $\beta = 0.25$ case).
7. **Parameterize it:** Click the small blank square next to the 115 value. Name the parameter `Branch_Dia`.

2.1.6 Extrude the Branch

1. Switch back to the **Modeling** tab, select `Sketch2`, and click **Extrude**.
2. In the Details View:

- **Operation: Add Material** — *this melts the pipes together.*
- **Direction: Normal**.
- **Extent Type: Fixed**.
- **Depth: 1500**. — *we use 1500 mm so it's long enough to clear the main pipe even if you tilt it to extreme angles like 30° later.*

3. **Parameterize the Depth:** In the Extrude details panel, click the small square next to the Depth value to create a new parameter. Name it `Branch_Extrude_Depth`.
4. Click **Generate**.

Defining the Extrude Depth Expression

In the **Parameter Set**, define `Branch_Extrude_Depth` using the following expression:

$$\texttt{Branch_Extrude_Depth} = 10 \times \texttt{Branch_Dia} + \frac{\texttt{Branch_Dia}\,/\,2}{\sin(\texttt{Inj_Angle})} \tag{2.1}$$

The two terms serve distinct purposes:

- $10 \times \texttt{Branch_Dia}$ ensures a minimum of **10 diameters** of straight pipe for flow development.
- $\frac{\texttt{Branch_Dia}\,/\,2}{\sin(\texttt{Inj_Angle})}$ is the additional distance needed for the branch's outer wall to clear the main pipe when tilted. For a vertical branch ($\theta = 90°$, $\sin = 1$), the extra is just the radius. For a shallow angle ($\theta = 30°$, $\sin = 0.5$), the extra is $2\times$ the radius, ensuring the extrusion fully exits the main pipe.

2.1.7 The "Fluid" Fix & Cleanup

1. Look at the bottom of your Tree Outline. Expand the **1 Part, 1 Body** folder.
2. Right-click the Solid and select **Rename**. Name it `T_Junction_Fluid`.
3. **CRITICAL STEP:** Click on `T_Junction_Fluid`. Look down at the Details View. Find the **Fluid/Solid** row. Click the drop-down and change it from `Solid` to **Fluid**.
4. Save your project and close DesignModeler.

2.2 Verifying the Setup

When you return to your main Workbench screen, look below your `Top_Injection_Base` system. You will now see a **Parameter Set** bar.

If you double-click that Parameter Set bar, you will open your master table. You will see columns for `Inj_Angle` and `Branch_Dia`. You can now type in any combination of angles and diameters you want to test in the rows below, and ANSYS will mathematically distort the geometry for you!

Chapter 3

Parametric Mesh Generation

This chapter details the robust, automated meshing strategy required for the hydrogen T-junction Detached Eddy Simulation (DDES). The setup utilizes a Body of Influence (BOI) to ensure the high-resolution mixing zone automatically scales with geometric parameter changes, alongside a microscopic inflation layer to capture critical near-wall hydrogen concentrations.

3.1 Pre-Requisite: Geometry BOI Setup

Before entering the Meshing application, the refinement zone must be defined in DesignModeler as a physical, parameterized body.

1. In DesignModeler, navigate to **Create** → **Primitives** → **Sphere**.
2. Set the **Origin Z** to **4600 mm** to center it over the junction.
3. Change the **Operation** to **Add Frozen** (Crucial: this prevents the sphere from merging with the pipe geometry).
4. Enter the baseline **Radius** (e.g., 500 mm) and click the box next to it to parameterize it as `BOI_Radius`.
5. Generate the sphere, rename the new body to `BOI_Zone`, and ensure its material is set to **Fluid**.

3.2 Global Setup & Quality Controls

These global settings ensure the mesh transitions smoothly across the fluid domain without overloading computational memory.

1. In Workbench, double-click the **Mesh** cell to open the application.
2. In the Outline tree, click on **Mesh**.

3. In the Details panel, set **Physics Preference** to `CFD` and **Solver Preference** to `Fluent`.
4. Under the **Sizing** section, apply the following:
   - **Element Size:** 15 mm
   - **Growth Rate:** 1.15
   - **Capture Curvature:** Yes
   - **Curvature Normal Angle:** 15° (ensures pipe walls remain perfectly circular)
5. Under **Advanced** (or Defaults), verify that **Element Order** is set to **Linear**.

3.3 Boundary Labels (Named Selections)

These labels define the physical boundaries for the Fluent solver.

1. **`main_inlet`:** Select the upstream circular face ($Z = 0$) → Right-click → **Create Named Selection**.
2. **`branch_inlet`:** Select the top circular face of the injection branch → Right-click → **Create Named Selection**.
3. **`outlet`:** Select the downstream circular face ($Z = 9200\,\text{mm}$) → Right-click → **Create Named Selection**.
4. **`wall`:** Select all remaining outer faces (main cylinder, branch cylinder, and the intersection corner) → Right-click → **Create Named Selection**.

3.4 Structured Core Algorithm

To maintain a highly organized core mesh, the MultiZone method is used.

1. Right-click **Mesh** → **Insert** → **Method**.
2. **Geometry:** Select the `T_Junction_Fluid` pipe body and click **Apply**.
3. **Method:** Select **MultiZone**.
4. **Src/Trg Selection:** Leave set to **Automatic**.
5. **Free Mesh Type:** Set to **Hexa Core**.

Note on Parametric Sweeps: If the MultiZone algorithm fails on extreme injection angles (e.g., 30°) during automated solving, the fallback strategy is to change this Method to **Tetrahedrons**, and subsequently use Fluent's Polyhedral conversion tool to optimize the cell count.

3.5 Parametric Wake Refinement (Body of Influence)

This step explicitly maps the high-resolution 5 mm cells to the `BOI_Zone` created in DesignModeler.

1. Right-click **Mesh** → **Insert** → **Sizing**.
2. **Geometry:** Select the `T_Junction_Fluid` pipe body and click **Apply**.
3. **Type:** Change to **Body of Influence**.
4. **Bodies of Influence:** Click the yellow selection box, select the `BOI_Zone` ghost sphere, and click **Apply**.
5. **Element Size:** Enter **5 mm**.

3.6 DDES Boundary Layer (Inflation)

A highly resolved boundary layer is strictly required for the Spalart–Allmaras DDES model to accurately predict near-wall hydrogen stratification.

1. Right-click **Mesh** → **Insert** → **Inflation**.
2. **Geometry:** Select the `T_Junction_Fluid` pipe body and click **Apply**.
3. **Boundary Scoping Method:** Change to **Named Selections**.
4. **Boundary:** Select the `wall` Named Selection.
5. **Inflation Option:** Change to **First Layer Thickness**.
6. **First Layer Height:** Set to **0.001 mm** (1 micron). This ensures a $y^+ \approx 1$ at a 6.9 MPa operating pressure.
7. **Maximum Layers:** Set to **15**.
8. **Growth Rate:** Set to **1.15**.

3.7 Automated Scaling Setup (Workbench Parameter Set)

To finalize the parametric automation, the Body of Influence must be mathematically linked to the branch diameter so it scales appropriately across all design points.

1. Save and close the Meshing application.
2. On the main Workbench screen, double-click the **Parameter Set** bar.
3. Locate the row for the `BOI_Radius` parameter.
4. In the **Expression** properties box, input the mathematical link to the branch diameter (e.g., if Branch Diameter is parameter `P2`, type:

   ```
   P2 * 4.3
   ```

5. Ensure the **Design Point Update Process** is set to **Update Full Project**.

Chapter 4

Literature Review

4.1 The Foundation and The Problem

4.1.1 The Big Picture Goal

To fight climate change, there is a major drive to reduce carbon dioxide emissions, especially from domestic heating.

The proposed solution is to create a *"blend"* by injecting **green hydrogen** into existing natural gas pipelines, targeting a hydrogen concentration of **5% to 20% by volume**.

At these percentages, consumers wouldn't need to change their home boilers.

4.1.2 The Core Engineering Challenge

Currently, the UK gas network is primarily made of steel and only allows a maximum hydrogen concentration of **0.1%**.

This limit must be safely raised to deliver the new hydrogen blends.

However, blending hydrogen into natural gas creates two directly conflicting problems:

The Energy Density Problem:
Pure hydrogen holds a lot of energy per kilogram, but its density is **9 times lower** than natural gas. This means a hydrogen-blended gas contains less energy per unit of volume. To deliver the same amount of heating energy to a home, you must either pump the gas faster or increase the pipeline pressure.

The Material Safety Problem:
Hydrogen causes carbon steel to become brittle—a process known as **embrittlement**. This reduces the steel's "toughness," making it much easier for existing microscopic defects or welds to turn into critical, fast-growing cracks under stress. Because of this danger, safety codes typically require that you *decrease* the pipeline pressure.

The Conflict:

Energy demand requires **higher** pressures, but safety regulations require **lower** pressures.

4.1.3 The Mixing Problem (The Focus of the Paper)

Hydrogen is typically injected into the main natural gas line through a **"T-junction"** (a pipe connecting at a 90-degree angle).

Because hydrogen is so light compared to natural gas, it lacks the momentum to penetrate deeply into the main gas flow.

Instead of mixing smoothly, the buoyant hydrogen floats to the top of the pipe. This is called **stratification**.

The authors' computer simulations show that injecting hydrogen from the top (**top-side injection**) results in poor mixing, leaving extremely high concentrations of hydrogen (**over 40%**) clinging to the upper wall of the pipe for long distances.

This is dangerous because exposing the top of the steel pipe to such high concentrations of hydrogen rapidly accelerates the **embrittlement** process.

Key Finding previewed in the abstract: Injecting the hydrogen from the bottom (**under-side injection**) forces the light hydrogen to rise *through* the natural gas, which promotes much better mixing and reduces the dangerous concentration on the pipe walls.

4.2 Model Description and Mathematical Framework

4.2.1 The Physical Setup

To make the simulation realistic, the authors based their model on the standard conditions of the UK natural gas network:

The Main Pipe (Natural Gas):

Modeled as an 18-inch (460 mm) diameter pipe operating at a pressure of 6.9 MPa (p_1). The gas flows at a speed of 10 m/s (u_1).

Temperatures:

Because pipes are buried underground, the natural gas (methane) temperature is set to 284 K (11.2 °C). The injected hydrogen is slightly warmer at 293 K (20 °C).

The Gases:

Natural gas is mostly methane, so the main flow is modeled purely as methane (CH_4)

with a molecular mass of 16 $\mathrm{g\,mol^{-1}}$. Hydrogen (H_2) has a molecular mass of 2 $\mathrm{g\,mol^{-1}}$ (meaning methane is 8 times heavier).

The Goal:

Achieve an average hydrogen volume concentration (V_R) between **5%** and **20%** downstream.

4.2.2 The Core Equations (Mixing & Concentration)

To figure out how much hydrogen to inject to hit that 5–20% target, the authors use the laws of conservation (mass and species).

Density

Using the Ideal Gas Law, the density (ρ) of the gas streams at the inlets depends on pressure (p), temperature (T), and molecular mass (M).

$$\rho_1 = \frac{M_{\mathrm{CH_4}}\, p_1}{R_g\, T_1} \qquad \text{(Mainline Methane)} \tag{4.1}$$

$$\rho_2 = \frac{M_{\mathrm{H_2}}\, p_2}{R_g\, T_2} \qquad \text{(Branch Hydrogen)} \tag{4.2}$$

Conservation Equations

The total mass going into the T-junction must equal the total mass coming out (Eq 4). Similarly, the mass of pure hydrogen injected must equal the mass of hydrogen in the final blended mix (Eq 5).

Volumetric Dilution ($\mathcal{D}$)

This is a crucial concept in the paper. It represents the ratio of the volume of gas injected from the branch to the volume flowing in the main pipe (Eq 7):

$$\mathcal{D} = \left(\frac{d_2}{d_1}\right)^2 \frac{u_2}{u_1} \tag{4.3}$$

To hit their target blends, they tested dilution factors of $\mathcal{D} = 0.05,\ 0.1,\ 0.2$, and 0.3.

Momentum Ratio (M_R)

This equation dictates how hard the hydrogen punches into the methane stream (Eq 8):

$$M_R = \frac{\rho_2 \, u_2^2}{\rho_1 \, u_1^2} \tag{4.4}$$

Why this matters: Because hydrogen is so light, the ratio of its density to methane's density is about 1/8. Therefore, if $M_R < 1$, the hydrogen jet has very weak penetration and will likely be swept away instantly by the heavy, fast-moving methane.

4.2.3 The Computational Math (Fluid Dynamics)

To track exactly where every particle of gas goes, the authors use standard fluid dynamics equations solved by a computer. You don't need to memorize these, but it is important to know what the computer is calculating:

Conservation of Momentum (Eq 9):
This is a version of the Navier-Stokes equations. It calculates how the velocity and pressure of the gas change, crucially including a gravity/buoyancy term (f_i) which drives the hydrogen upward.

Species Advection (Eq 10):
This tracks the physical movement and spreading (mass fraction, Y) of the specific hydrogen and methane molecules through the pipe.

Energy/Enthalpy (Eq 11):
This tracks heat and kinetic energy. While temperature differences are minor here, the computer still tracks them to ensure total physical accuracy.

4.2.4 Implementation and Validation

The Software:
They used an open-source computational fluid dynamics (CFD) software called **OpenFOAM**, specifically a solver named `rhoReactingBuoyantFoam` (with chemical reactions turned off, since the gases are just mixing, not burning).

Turbulence Model:
Because gas flowing at 10 m/s is highly turbulent, they used a "Large-eddy simulation" (LES) approach, specifically the **Spalart–Allmaras DES** model. This is considered state-of-the-art for calculating turbulence near pipe walls.

Validation:
To prove their computer model was accurate before testing the hydrogen, they ran a simulation of a well-documented physical experiment (mixing hot and cold water in a T-junction). Their computer model matched the real-world physical data very well

(shown in Figure 2), giving confidence that their hydrogen simulations would also be accurate.

4.3 Governing Equations and Numerical Methods

4.3.1 The Governing Equations (The Laws of Physics)

The model relies on three fundamental differential equations that describe how fluids move and exchange energy.

Equation 9: Conservation of Momentum (Navier–Stokes)

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu_e\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + \rho f_i \tag{4.5}$$

What it means: This is essentially Newton's Second Law ($F = ma$) applied to a fluid. It calculates how the velocity of the gas (u) changes over time (t) and space (x).

Key terms: The flow is driven by changes in pressure (p), slowed down by viscous friction (μ_e), and affected by gravity (ρf_i). **The gravity term is incredibly important here** because hydrogen's extreme lightness means buoyancy forces will try to push it to the top of the pipe.

Equation 10: Species Conservation

$$\frac{\partial(\rho Y^*)}{\partial t} + \nabla \cdot (\rho Y^* u) = \nabla \cdot (\mu_e \nabla Y^*) \tag{4.6}$$

What it means: This tracks the actual mixing. Y^* represents the mass fraction of either hydrogen or methane. The left side calculates how the gas is carried along by the main flow (advection), while the right side calculates how it spreads out into the surrounding gas (diffusion).

Equation 11: Energy Equation

This equation tracks enthalpy (h) and kinetic energy (K) to figure out if the gases heat up or cool down as they mix. Because the temperature variations in this specific pipeline scenario are very small, they use constant "heat capacities" (c_p) for the hydrogen and methane to simplify the math without losing accuracy.

4.3.2 Turbulence Modeling (Taming the Chaos)

Because the gases are moving fast (high Reynolds number), the flow is highly turbulent—full of chaotic, swirling eddies. Simulating every single microscopic swirl requires too much computing power.

The Solution:

Based on past OECD (Organisation for Economic Co-operation and Development) benchmark tests, the authors chose the **Spalart–Allmaras DES (Detached Eddy Simulation)** model.

How it works:

Near the pipe walls, it uses mathematical averages to guess the turbulence. But in the middle of the flow (the "interior"), it switches to Large-Eddy Simulation (LES) to actually calculate the large swirls that are responsible for pushing the hydrogen and methane together.

4.3.3 Boundary Conditions (Setting the Stage)

To make the simulation realistic, you have to define how the gas enters the pipe.

Fluid in a pipe doesn't move as a solid flat block. Friction against the steel walls makes the gas at the edges move slower than the gas in the dead center. The researchers use a **"1/7th power-law"** ($n = 7$) equation to ensure the virtual gas entering their simulation has a realistic, slightly bullet-shaped velocity profile.

4.3.4 Implementation and Validation (Building the Virtual Pipe)

Finally, they explain *how* they built this in a computer and proved it was accurate.

The Software:

They used **OpenFOAM**, a famous open-source CFD program, specifically using a solver called `rhoReactingBuoyantFoam`. This solver is designed to handle gases with drastic density differences (buoyancy), making it perfect for the hydrogen/methane problem.

The Mesh:

They chopped the 3D virtual pipe into up to **4.8 million tiny 3D blocks (cells)**. They packed these cells very tightly together near the T-junction and along the pipe walls to get ultra-high-resolution data exactly where embrittlement is a risk.

Validation:

Before trusting their hydrogen model, they ran a simulation of a well-documented

physical experiment (water mixing in a T-junction). By matching a fluid dynamics metric called the **Froude number** (which balances the momentum of a fluid against gravity/buoyancy), they proved their OpenFOAM setup correctly predicted real-world lab results. Only after passing this test did they move on to the hydrogen simulation.